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Foreign Manufacturing, Regional Agglomeration and Technical Efficiency in UK Industries: A Stochastic Production Frontier Approach

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DRIFFIELD N. and MUNDAY M. (2001) Foreign manufacturing, regional agglomeration and technical efficiency in UK industries: a stochastic production frontier approach, *Reg. Studies* 35, 391–399. The paper examines how far foreign manufacturing investment in UK industries, together with the spatial agglomeration of those industries, affect technical efficiency. The paper links research on the estimation of technical efficiency, with those literatures demonstrating the economies associated with foreign direct investment and spatial agglomeration. The methodology involves estimation of a stochastic production frontier with random components associated with industry technical inefficiency, and a standard error. The paper also explores whether the degree of foreign involvement has a greater impact on technical efficiency where the domestic industry sector is characterized by comparatively high productivity and spatial agglomeration. The policy implications of the analysis are discussed.

Foreign manufacturing

Regional industry agglomeration

Industry efficiency

DRIFFIELD N. et MUNDAY M. (2001) L'industrie étrangère, l'agglomération régionale et l'efficacité technique des industries situées au Royaume-Uni: une façon portant sur la frontière de la production stochastique, *Reg. Studies* 35, 391–399. Cet article cherche à examiner jusqu'à quel point l'investissement industriel au Royaume-Uni, conjointement avec l'agglomération géographique des industries en question, a une incidence sur l'efficacité technique. L'article relie la recherche sur l'estimation de l'efficacité technique à la documentation qui démontrent les économies dues à l'investissement direct étranger et à l'agglomération géographique. La méthodologie nécessite une estimation de la frontière de la production stochastique, avec des composantes aléatoires associées au manque d'organisation industrielle technique, et une erreur-type. L'article cherche aussi à examiner si, oui ou non, le taux d'investissement étranger a plus d'incidence sur l'efficacité technique là où l'industrie nationale se caractérise par une productivité et une agglomération géographique relativement élevées. Les retombées de cette analyse pour la politique se voient discuter.

Industrie étrangère Agglomération industrielle régionale
Efficacité industrielle

DRIFFIELD N. und MUNDAY M. (2001) Produktion ausländischer Firmen, regionale Ballung und technische Leistung in der Industrie des Vereinigten Königreichs: ein stochastischer Produktionsgrenzenansatz, *Reg. Studies* 35, 391–399. Dieser Aufsatz untersucht, wie Investitionen ausländischer Hersteller in der Industrie des Vereinigten Königreichs sich bei gleichzeitiger räumlicher Ballung dieser Industrien auf technische Leistung auswirken. Der Aufsatz verbindet Untersuchungen der Schätzung technischer Leistung mit Veröffentlichungen, welche sich mit der Darstellung von Wirtschaften beschäftigen, die mit Direktinvestitionen des Auslands und räumlicher Ballung in Verbindung gebracht werden. Bei dieser Methodik werden Schätzungen der stochastischen Herstellungsgrenze mit Zufallskomponenten einbezogen, die mit technischer Inkompetenz der Industrie in Verbindung gebracht werden, sowie einem Standardfehler. Darüberhinaus untersucht der Aufsatz auch, ob das Ausmaß ausländischer Mitarbeit sich stärker auf technische Leistung auswirkt wo der einheimische Industriesektor sich durch verhältnismäßig hohe Leistung und räumliche Ballung auszeichnet. Es werden Implikationen der Analyse für Zielsetzungen diskutiert.

Produktion ausländischer Firmen
Regionale Industrieballung Industrielle Leistung

INTRODUCTION

This paper uses three digit industry data from the UK Census of Production for the years 1984–92 to establish the determinants of UK industry technical efficiency,

and the role played by both industry levels of foreign manufacturing investment, and the spatial agglomeration of similar industry activities. Having established these relationships, the paper then explores how far the impacts of foreign firm spillovers vary according to

existing levels of industry productivity and spatial agglomeration. The method adopted involves the estimation of a stochastic production frontier with random components associated with industry technical inefficiency and a standard error. The contribution then attempts to link research on the estimation of technical efficiency with research that has demonstrated the economies associated with foreign direct investment and spatial agglomeration.

The remainder of the paper is divided into four sections. The second section of the paper outlines the significance of foreign manufacturing industry to the UK economy, and particularly its contribution to regional economic prospects. The section then explores the linkages between foreign manufacturing, regional agglomeration economies and the performance of domestic industry sectors. The third section considers the measurement of industry technical efficiency, and the methodology employed in estimating a stochastic production frontier. The section then specifies a model to assess the role of regional industry agglomeration and foreign industry spillovers on industry technical efficiency, and discusses the data sources and variables used in the modelling process. The fourth section discusses the results from the model estimation. The final section contains conclusions, focusing on the ramifications of the results for UK policy on inward investment in its regions.

FOREIGN MANUFACTURING, REGIONAL INDUSTRY AGGLOMERATION AND INDUSTRY PERFORMANCE

The UK continues to lead other areas of the European Union (EU) as a host for inward investment. The UK inward foreign direct investment stock was estimated at \$327 bn in 1998, just over one-fifth of the total EU stock of direct investment, and nearly \$100 bn greater than that in the next largest host – Germany (UNCTAD, 1999). Total foreign direct investment inflows into the UK were ECU56.4 bn in 1998. Around 35% of this investment was from Europe, and 57% from North America. Inward investment in manufacturing only represents parts of these totals. For example, in 1998, around 50% of investment inflows were in financial and business services, with under 30% in manufacturing industries. Mergers and acquisitions have represented a growing proportion of inflows in the recent past (UNCTAD, 1999; EUROSTAT, 1999).

The most explicit impact of foreign manufacturing in the UK has been the creation of new employment, often in regions damaged by structural decline. Many of the periphery regions of the UK feature large concentrations of foreign capital – particularly Wales, Scotland, Northern Ireland and Northern England (see COLLIS, 1992; HILL and MUNDAY, 1994). However, the value of the economic contribution of multinational

firms in these regions has been questioned. For example, critical research has focused on the scale of input–output linkages between multinational firms and locally based suppliers, the quality of employment provision, the production-only nature of regional operations and more general issues of embeddedness (YOUNG *et al.*, 1988; YOUNG *et al.*, 1994; TUROK, 1993; CHRISTODOULOU, 1996; LOVERING, 1998). Concern on the economic contribution of multinational firms in the UK regions also needs to be considered in the contexts of: falling real levels of resources for location marketing in the UK periphery; potentially wasteful interregional battles to attract the largest inward investment projects; and several well publicized failures amongst heavily subsidized inward investors in the UK periphery during 1998–99, including Siemens in Northern England, Hyundai in Scotland and LG Semicon in Wales. Recent, and large, inward investor failures have served to heighten debate on levels of public funding for inward investors, institutional roles and efficient land use (see for example, PHELPS *et al.*, 1998), and have brought into sharper relief the opportunity costs associated with inward investment attraction, particularly in terms of the development of indigenous enterprise.

The continuing debate on the role of multinationals in regional and national development processes has prompted policy orientated research urging UK regional development organizations to be more selective in their targeting of foreign capital (BRAND *et al.*, 2000). Linked to this need for targeting is a growing recognition, following PORTER, 1996, that a regional policy which actively pushes foreign investors to areas bereft of appropriate social and physical infrastructures, and agglomerations of like investment, could impede the development of competitive clusters, with longer term ramifications for national competitiveness (see also DRIFFIELD and MUNDAY, 2000).

Clearly, the above concerns on the value and location of foreign capital in the UK, together with the nature of present and future policy interventions, need to be assessed within a wider research framework that demonstrates the significance of multinational investment and general industry agglomeration for domestic industry competitiveness. The process of foreign direct investment, as opposed to licensing knowledge to a domestic firm, partially reflects an internalization of proprietary knowledge. DUNNING, 1985, 1988, demonstrates that the ownership advantage of foreign firms (which potentially gives them a competitive edge over domestic firms) can be in terms of both tangible and intangible assets. However, several categories of ownership advantages have the characteristics of a public good – particularly in the realms of technology. As such the ownership advantage of an incoming multinational may be transitory, with some potential for new technologies to spillover and be appropriated into the host sector, with resulting improvements in

domestic industry competitiveness. For example, BLOMSTRÖM and KOKKO, 1996, in a review of the significance of multinational enterprise spillovers, show that multinational enterprises can improve allocative efficiency through entering industries with high entry barriers, and can potentially break down domestic monopolies and improve competition. Moreover, the presence of multinational enterprises can improve rates of technical diffusion, remove supply bottlenecks and prompt the introduction of new operational techniques through, for example, buyer–supplier linkages (for a further review see DUNNING, 1993).

The measurement of the significance of spillovers from multinationals to the domestic sector remains difficult. DUNNING, 1993, provides a review of studies which have explored the relationship between host industry performance and foreign investment at the industry and aggregate level. Much of the evidence reveals a positive association between foreign involvement and indigenous performance – particularly when this latter is assessed in terms of revealed comparative advantage. In the UK, qualitative evidence of the development of improved operational practices as a result of the presence of multinational enterprise has been widespread – particularly in the case of Japanese manufacturing investment, which has had wide ranging impacts on the operations of suppliers and competitors in realms of problem solving, manufacturing management, personnel management techniques and industrial relations practice (see OLIVER and WILKINSON, 1992). Moreover, the superior productivity of multinational enterprise over its UK counterparts has been established empirically in several studies – thus providing an essential foundation for spillovers to occur (see for example, DAVIES and LYONS, 1992; OULTON, 1999). However, empirical evidence of industry level spillovers is more scarce. BARRELL and PAIN, 1999 (see also BARRELL and PAIN, 1997; DRIFFIELD, 2001) analysed the impact of inward investment on technical progress across the UK, France, Germany and the Netherlands, and found significant effects from foreign direct investment. A 1% rise in the real stock of foreign direct investment (FDI) was estimated to raise the level of technical progress by 0.18% in these economies. Turning to the UK specifically, HUBERT and PAIN, 1999, explored the impact of sector-specific inward investment on labour productivity and technical progress, and found evidence of spillovers in the manufacturing sector, with smaller yet significant effects in the distribution and business services sectors. Hubert and Pain also found evidence across 15 manufacturing sectors that the presence of multinational enterprise had affected technical progress both within and between these industry groups. Importantly, this study, by adopting a disaggregated industry approach, allowed for the fact that the influence of FDI spillovers was likely to vary between broad industry classifications. DRIFFIELD and MUNDAY, 2000, examined the

relationship between the comparative advantage of UK industries, and new inward investment into these industries. They demonstrated that the extent of foreign manufacturing investment in an industry was a significant determinant of industry comparative advantage, and that industry comparative advantage itself, together with a series of industry specific characteristics, were important determining factors in explaining new foreign manufacturing investment, thus providing evidence of the dynamic benefits of FDI spillovers in the UK economy and its regions.

The potential for FDI-induced industry level productivity spillovers and resulting improvements in industry efficiency may also be linked to other characteristics of the indigenous industry sector. WANG and BLOMSTRÖM, 1992, for example, suggest that spillover potential is positively associated with the size of the technology gap between the foreign and domestic sector. WAKELIN *et al.*, 1999, also explored how far the skills mix of the work-force, competitiveness of the market place and the magnitude of the productivity gap might influence the propensity of domestic firms to benefit from foreign investment induced spillovers. They demonstrated that the impact of FDI on productivity increased as competition and industry skill levels grew. However, larger technology gaps were shown to hinder productivity spillovers.

In summary, there is some expectation from the reviewed literature that foreign manufacturing within an industry sector is connected to improvements in the technical efficiency of domestic firms in that sector. There is the additional expectation that the impact of foreign firms on domestic industry technical efficiency is related to industry specific characteristics. In particular, and following the discussion above, that the impact of foreign investment on domestic sectors in the UK is influenced by the extent of the technology gap, with the expectation that domestic firms in comparatively productive sectors are best able to assimilate the technology employed in the more productive foreign sector. Conversely, while low productivity industries may attract FDI, it is likely that such investments are characterized by lower levels of technology than investments into high productivity industries, or that the technology gap will be too large for domestic firms to assimilate the foreign technology. In either case, the scope for spillovers in low productivity industries is expected to be limited.

Another industry level characteristic that could affect both the domestic industry technical efficiency, and the propensity for the domestic sector to benefit from foreign investment induced spillovers is industry spatial agglomeration. Agglomeration externalities are an increasing function of the number and size of proximate firms. They may arise because of specialized local markets for labour and intermediate products. Moreover, physical proximity may aid the process of inter-firm knowledge spillovers, for example, making casual

communication less costly in cases where firms are unable to prevent externalities from scientific and managerial development. The agglomeration of industry activity may impact on productivity growth because of its influence on the rate of technical change (BEE-SON, 1987). In particular the spatial concentration of transaction-linked sectors can determine the speed with which new knowledge is developed, the speed of dissemination and the speed with which new knowledge and production techniques are incorporated into industry processes. A key area of debate, however, is the distance over which such agglomeration benefits are significant (KRUGMAN and VENABLES, 1995; AUDRETSCH, 1998).

The concept of agglomeration is linked with new approaches in economic geography which have highlighted the competitive potential associated with tight demand and supply interlinkages among regional clusters of allied industries (see, for example, SCOTT, 1988; PORTER, 1990). In the UK, more attention is now being given to ways in which the embeddedness of existing groups of inward investors can be improved and agglomeration externalities developed (TUROK, 1993; DEPARTMENT OF TRADE AND INDUSTRY, 1998; IZUSHI, 1999). Whilst industry agglomeration may play a role in reducing technical inefficiency in the domestic sector as a whole, there is also the possibility that industries that are regionally concentrated might also benefit most from foreign investment induced productivity spillovers, with geographical proximity expected to affect the degree of knowledge transmission through labour markets, buyer-supplier partnerships and general communities of interest. This does not mean that where industries are spatially dispersed that productivity spillovers will not occur, but that the scope for such spillovers is greater where industries are less dispersed.

In conclusion, the review suggests that both the level of foreign involvement within an industry sector, and the agglomeration of that industry, will impact positively on the technical efficiency of that industry, and that the impact of spillovers may be more pronounced in industries characterized by comparatively high levels of productivity, and spatial agglomeration. In the next section we review the measurement of industry technical efficiency, and the method adopted to explore the significance of foreign manufacturing investment and industry spatial agglomeration on domestic industry efficiency.

METHODOLOGICAL APPROACH TO INDUSTRY TECHNICAL EFFICIENCY AND MODEL SPECIFICATION

As discussed above, this paper seeks to determine the extent to which productivity spillovers from the foreign sector and agglomeration externalities significantly impact on domestic industry technical efficiency. A

frontier approach to industry efficiency is adopted here. Approaches involving the estimation of stochastic frontiers were pioneered by AIGNER *et al.*, 1977; and MEEUSEN and VAN DEN BROECK, 1977. The underlying rationale of this methodological approach is that it is possible to analyse technical efficiency in terms of observed deviations from an optimal production frontier. The aggregate production process is therefore modelled as a single-output production frontier, that is, the model effectively imposes an identical production function across all industries in the sample.¹ A further discussion of this methodology and the specific treatment of industry inefficiency is found in the Appendix and GREENE, 1993.

Within the framework outlined above, and within the context of the literature review, the basic hypotheses are that the degree of foreign involvement in a defined industry sector, and the regional agglomeration of that industry improve efficiency (or in terms of the frontier framework actually reduce inefficiency). Moreover, that the degree of foreign involvement has a greater impact on technical efficiency where the domestic industry sector is characterized by comparatively high productivity and spatial agglomeration. The approach adopted here therefore is to employ a Cobb-Douglas production function, and to then estimate this model within the frontier panel framework. Simply, the model takes the following form:

$$\ln(Q_{it}) = \beta_0 + \beta_1 \ln(K_{it}) + \beta_2 \ln(L_{it}) + \beta_3 \ln(NL_{it}) + \beta_4 \ln(RD_{it}) + V_{it} - U_{it} \quad (1)$$

where: Q is domestic industry gross value added; K is given as the capital stock of the domestic industry; L is employment of manual labour in the domestic industry; NL is employment of non-manual labour in the domestic industry; RD is the stock of R&D capital of the domestic industry; V_{it} are random errors; and U_{it} is the measure of technical inefficiency based on a random-effects specification (for further discussion of this specification see BATTESE and COELLI, 1995).

The model distinguishes between types of labour. Clearly, labour is not homogeneous, and as such, levels of manual (L) and non-manual labour (NL) may be expected to impact differently on output. The R&D stock of the industry is included with ample evidence following GRILICHES, 1979, that R&D impacts on output across industries. The appropriate specification is to explore technical inefficiency effects in terms of the variable under investigation where:

$$U_{it} = \delta_0 + \delta_1 \text{FOREIGN}_{it} + \delta_2 \text{AGGLOM}_{it} \quad (2)$$

Here $AGGLOM$ is the regional industry agglomeration variable, and $FOREIGN$ is the variable that shows the degree of foreign penetration of the given industry sector. The hypothesis to be tested is that the $FOREIGN$ and $AGGLOM$ variables are negative and significant, that is, that increased agglomeration

and FDI in an industry actually reduce technical inefficiency.

This approach has several advantages over the more traditional methods that seek to link FDI to total factor productivity growth (COE and HELPMAN, 1995). First, we do not seek to explain total factor productivity growth in terms of FDI, but allow total factor productivity to be determined from the production function, in terms of the efficient frontier for the industry. It is then possible to test for the stability of the coefficients from equation (1) across industries. We then seek to determine the efficiency frontier for each industry, and then explain the stochastic inefficiency in terms of equation (2)

The definitions of the variables used are summarized in Table 1. The data comprises three-digit industry information (UK Standard Industrial Classification 1992) taken from the UK Business Monitor series which reports the results of the Annual Census of Production undertaken by the Office of National Statistics (ONS). The ONS also provided corresponding data for the foreign-owned sector only, which allowed

for calculation of data for the domestic sector. Industry data were available for the years 1984–92, and after omitting various ‘miscellaneous’ industry categories, there were 102 industry observations for each year.

Foreign penetration at the industry level (*FOREIGN*) is measured as the share of industry value added that is accounted for by foreign owned firms. Agglomeration of industry was measured in the following way. The Census of Production reports the industry gross value added for each of the 11 UK standard planning regions of the UK. The regional shares of given industry value added were calculated, and then a location quotient was derived. The location quotient reveals how specialized a region is in terms of a given industry. The location quotient was calculated by dividing the region’s share of the selected industry UK gross value added, by the same region’s share of total UK manufacturing gross value added. A location quotient of greater than 1 indicates that the region in question has a share of selected industry gross value added greater than its size in terms of share of total UK manufacturing value added would suggest. For each of the 102 defined industries, this gave a series of 11 location quotients. The standard deviation of these location quotients was calculated, and then a coefficient of variation. The value of the coefficient of variation (*AGGLOM*) is the measure of agglomeration used here. Where shares of industry value added are evenly spread across regions then the regional location quotients would tend towards 1, and the resulting standard deviation (and coefficient of variation) would tend to 0. More spatially concentrated industries would tend to have higher coefficients of variation. It is important to recognize that this variable describes regional concentrations of industry activity, and is hence only a guide to the existence of clusters of allied industry activity. A more comprehensive measure of the agglomeration would take into account the scale of industry input–output linkages within a defined space. At present it is not possible to calculate such a variable across all UK regions. Moreover, industry activities in the UK do not respect planning region boundaries with some clusters of allied activities spanning defined administrative boundaries.

MODEL ESTIMATION AND RESULTS

As outlined above, the initial stage in the estimation of the frontier is to determine the appropriate specification for the frontier model. This involves a test based on technical efficiency restrictions implied by the different error structures (see BATTESE and COELLI, 1992; KUMBHAKAR, 1993). The results from this process confirmed that a Cobb–Douglas production function was an appropriate specification of the stochastic frontier.²

These results presented in Table 2 show that regional concentration and foreign share determine domestic

Table 1. Data and variables

All monetary variables were put into real terms (1989 prices). Industry data for 102 three-digit industry sectors (SIC 92) derived from the UK Census of Production. Data available for 1984–92.

Dependent variable

Q Domestic industry output: gross value added as defined in the Report on the UK Census of Production (mean £822·25 m; s.d. £1,017·31 m).

Independent variables

K The capital stock of the domestic industry, the change in this is given by net capital investment in the UK owned sector. Data on the capital stock are not available at this level of aggregation, so the sum of net investment over the previous five years is used as a proxy. A standard depreciation rate of 10% is used (mean £471·85 m; s.d. £508·27 m).

L Manual labour – employment (thousands) of operatives in domestic owned industry (mean 33·53; s.d. 39·96).

NL Non-manual labour is employment (thousands) of non-operatives in domestic owned industry (mean 17·87; s.d. 19·93).

RD R&D expenditure in the domestic industry sector. The stock of R&D is measured as the sum of R&D expenditure over the previous five years, depreciated at the standard rate of 10% p.a. This is available in Business Monitor MO14 (mean £141·71 m; s.d. £439·78 m).

Inefficiency parameters

FOREIGN percentage share of industry value added that is accounted for by foreign owned manufacturing firms (mean 16·87%; s.d. 12·58%).

AGGLOM Industry agglomeration measured as the coefficient of variation for industry level location quotients across the 11 standard planning regions of the UK for 1984–92 (mean 0·908; s.d. 0·560).

Table 2. Results of estimation of stochastic frontier

Variables	Coefficient	t-values
Dependent variable $\ln(Q)$		
Constant	3.637	19.32
$\ln(K)$	0.276	14.16
$\ln(L)$	0.164	5.30
$\ln(NL)$	0.461	15.78
$\ln(RD)$	0.077	8.34
Inefficiency parameters		
δ_0	0.560	2.97
FOREIGN	-0.459	10.05
AGGLOM	-0.515	2.02
Variance parameters		
γ	0.926	24.16
σ_s^2	0.161	15.17
Log L	-499.46	
One-sided residuals $\chi^2(4)$	6.111	($p = 0.191$)
N	918 (9 years)	

Notes: Further diagnostics: all variables were significant at the 99% level. The specification of this model passes the usual 'basket' of tests. The value of γ is close to 1, and significant, indicating that the specification is correct, and that it is appropriate to model FOREIGN and AGGLOM as technical efficiency parameters rather than as part of the production function. It is clear that a $\chi^2(3)$ test rejects the hypothesis that $\delta_0 = \delta_1 = \delta_2 = 0$ (test statistic = 13.45, critical value = 7.82).

industry level efficiency. Both coefficients are negative as expected, illustrating that FDI and regional concentration serve to move industries nearer to the technical efficiency frontier. As such, inward investment and agglomeration economies serve to improve productivity in UK manufacturing with expected advantages in terms of overall national competitiveness.

In order to further examine the relationship between technical efficiency, agglomeration and foreign penetration, equations (1) and (2) were re-estimated using a series of sub-samples of the data. First, the sample was split according to observed levels of industry labour productivity (i.e. gross value added over employment, with data also available from the UK Census of Production). Second, the sample was split according to observed levels of industry agglomeration, that is, on the basis of the AGGLOM variable outlined above. A simple likelihood ratio test revealed that the coefficients from equation (1) were consistent across the two sub-samples, such that the parameters from equation (1) can be treated as consistent across the two sub-samples.³ However, this was not the case for the determinants of technical inefficiency. A Chow test based on the inefficiency equation (2) rejected the hypothesis that the coefficients on the FOREIGN and AGGLOM variables were consistent across the sub-samples, based on productivity or agglomeration. This means, for example, that the effect of foreign involvement in a given industry sector could vary according to whether the industry is characterized by comparatively high or low productivity. Therefore the equation must be

Table 3. Inefficiency parameters for foreign variable: re-estimation on sub-samples

	High productivity		Low productivity	
	Coefficient	t-value	Coefficient	t-value
δ_0	0.701	0.233	1.548	0.519
FOREIGN	-0.087	-2.858	-0.013	-0.466
AGGLOM	-0.341	-6.982	-0.660	-9.005
	High agglomeration		Low agglomeration	
	Coefficient	t-value	Coefficient	t-value
δ_0	0.275	1.811	1.516	0.016
FOREIGN	-0.121	-2.572	-0.018	-0.603
AGGLOM	-0.394	-5.698	-0.654	-9.862

estimated separately for these sub-groups. The results from the re-estimation of equation (1) are found in Table 3.

These results demonstrate that the mechanisms suggested earlier in the paper are indeed important in explaining variations in total factor productivity. For example, spillovers from FDI are only significant in industries of above average regional concentration. This suggests that a critical level of regional concentration is a necessary condition for spillovers from FDI to occur. As such, subsidies designed to attract foreign manufacturing industry away from core regions into the periphery could lead to many potential externalities being lost. These results also suggest that spillovers from foreign manufacturing in low productivity industries are negligible. There are two possible explanations here. First, foreign investment attracted to low productivity sectors could be labour intensive, with levels of endemic technology, and as such technology spillovers are limited. Second, domestic firms in these low productivity industries are unable to assimilate superior foreign technology.

CONCLUSIONS

The results of this paper reveal something of the dynamic benefits of attracting FDI in the UK case. The extent of foreign ownership in a domestic industry sector is a determinant of technical efficiency. Such improvements in technical efficiency are expected to feed through into the international competitiveness for UK industry sectors. These findings are important in the context of concerns over the contribution of inward investment to national and regional development processes.

As expected, the role of foreign investment induced spillovers in improving technical efficiency does vary according to industry characteristics. The evidence in the paper suggests that spillovers are more pronounced in sectors that are relatively productive and regionally concentrated. One possible corollary of this is that where new inward investment contributes to the spatial agglomeration of similar facilities, then it may have a

role in improving regional productivity. Further research might then demonstrate how far the location pattern of foreign manufacturing investment across the UK regions has actually served to narrow or widen the productivity gap between these regions. That the impacts of foreign spillovers differs according to industry productivity and geography also has ramifications for UK policy. In particular, there may be greater long term returns from location marketing where foreign capital is attracted to UK sectors where revealed technology gaps are smaller. Inasmuch as these gaps might be assessed in terms of revealed comparative advantage, then marketing might be targeted on those UK industries where such advantage is growing. Similarly, given the significance of spatial agglomeration in determining the impacts of foreign spillovers, then policy might be orientated to encourage industry clusters. There is currently little evidence that UK (and European) measures to influence industry location take much explicit account of the importance of the agglomeration of allied activities. Rather, measures have focused on the correction of structural problems in periphery areas, with little regard for how far such intervention could rob industries of the advantages of proximity to allied activities. UK policy may be changing: the 1998 Department of Trade and Industry White Paper on Competitiveness stressed that the government will continue to focus on attracting inward investment, but that there will be a focus on attracting projects which support the development of strong sectoral or technology based clusters (DEPARTMENT OF TRADE AND INDUSTRY, 1998). The White Paper admits that clusters are unlikely to be created by public policy, but that government can at least encourage their formation and growth. The findings of this paper would suggest that this may be a valuable development, although potentially difficult to follow through in an environment of stiff interregional competition to attract foreign manufacturing projects.

NOTES

1. The extent to which it is appropriate to impose an identical production function across all industries in the sample is essentially an econometric issue. Imposing an 'average' production function across 102 industries is a concern (see also BEESON and HUSTED, 1989). Whilst one cannot (beyond allowing for fixed effects) test for n different sets of coefficients, it is possible to test for differences by splitting the sample according to groups that may be considered technologically similar. This was tested directly by determining whether this restriction is rejected using a Chow-type test, when the sample is split into various sub-samples. The estimation technique used here however is maximum likelihood, rather than least squares, so a test based on the likelihood functions, rather than the residuals is appropriate. In the cases presented here, the restriction of identical parameters is not rejected (see also note 2).
2. The appropriate specification of production function was determined empirically. The most rigorous approach to this is to start with an unrestricted translog production function, and then to test the joint restriction that the cross product and squared terms may be omitted using a standard F-test. With a fixed effects model this involves an $F(k, n-k)$ test, where k represents the number of terms potentially omitted from the model, and n is the number of cross sections in the data. The test statistics here generated a value of 1.453 ($p=0.1703$) such that the restriction imposing a Cobb-Douglas model on these data cannot be rejected. When the exercise was completed using the unrestricted translog form, the results regarding the significance of foreign industry penetration and regional concentration did not change. Tests also confirm that any restriction that $\beta_1 = \beta_2$ (equation (1), is invalid and that, as such, labour is heterogeneous in the production function and should be treated as such. The stock of R & D capital in the industry was also a significant determinant of the frontier. The results of the estimation of these models are presented below.
3. This is similar to the chi-squared test outlined in note 1.

APPENDIX

Frontier approaches to industry technical efficiency

The underlying rationale of the frontier approach is that one can analyse technical efficiency in terms of observed deviations from an optimal production frontier. The aggregate production process is therefore modelled as a single-output production frontier (GREENE, 1993; AIGNER *et al.*, 1977; MEEUSEN and VAN DEN BROECK, 1977). The estimation of a stochastic frontier involves the implicit assumption that there exist two components to inefficiency. The first of these is associated with technical inefficiency, and the second with standard random error.

While stochastic frontier analysis can be carried out using cross section data, the assumptions concerning the firm specific inefficiency are much more restrictive than if panel data can be employed. Following BATTESE and COELLI, 1995, the standard stochastic production frontier employs the following model:

$$Y_{it} = \exp(x_{it}\beta + v_{it} - u_{it}) \quad (3)$$

where: Y_{it} denotes the production of unit i at time t ; x_{it} are the independent variables, the inputs into the production unit; and β is a vector of unknown parameters to be estimated. v_{it} are the random errors, assumed to be independently identically distributed $(0, \sigma_v^2)$, and independently distributed of u_{it} . u_{it} are a set of non-negative random variables, associated with the technical inefficiency of production, assumed to be independently distributed, such that u_{it} is obtained by truncation at zero of the normal distribution – the so-called 'half-normal' distribution.

$$f(u) = (1/\sigma_u)\phi[(u - \mu)/\sigma_u]/\Phi[\mu/\sigma_u]$$

$$E(u) = \mu + \sigma_u\phi[\mu/\sigma_u]/\Phi[\mu/\sigma_u] = \mu + \sigma_u\lambda_u$$

$$\text{Var}(u) = \sigma_u^2[1 - \lambda_u(\mu\sigma_u + \lambda_u)]$$

There are then numerous approaches to the treatment of this inefficiency term. For example, GATHON and PERELMAN, 1992, assume that U_{it} can be specified as U_i , and estimated as a time invariant firm specific inefficiency within a random or fixed effects framework. CORNWELL *et al.*, 1990, however, show that one may wish to relax the assumption of fixed inefficiencies, and specify a model in the following form:

$$U_{it} = U_i + r_i \quad (4)$$

assuming that a similar effect impacts on each unit within the same period.

However, BATTESE and COELLI, 1995, propose a maximum likelihood approach for simultaneous estimation of these parameters, for both the stochastic frontier, and the technical efficiency effects. The likelihood function is expressed in terms of the variance parameters:

$$\sigma_\epsilon^2 \equiv \sigma_v^2 + \sigma^2 \quad \text{and} \quad \gamma \equiv \sigma^2 / \sigma_\epsilon^2$$

This may be estimated using the *Frontier 4.1* package, see COELLI, 1996.

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